

Error redirection

In error redirection, three files are usually open whenever any program or command is executed at the terminal. These files are standard output, standard input and standard error. Whenever the program is run, these files are presented, and file descriptors of all these files get associated with them.

While redirecting, every stream has its own commands. Single or double brackets are used to redirect standard output. If a file doesn't exist then a new file is generated. Commands having a single bracket can overwrite the existing content. To redirect standard error, you have to write the command '2>'.

Piping

The pipeline consists of single or multiple commands, separated by the pipe symbol as (`|`). The shell connects the standard output preceding this symbol to the standard input following the pipe symbol. The pipeline also has an effect similar to redirecting the standard output of a single command to any file and then using the same file as standard output for another command. Syntax:

```
command_a [arguments] | command_b [arguments]
```

Pipeline is used for the president command to affect the three commands-

- `command_a [arguments] > temp`
- `command_b [arguments] <temp`
- `rm temp`

The most powerful asset of input and output redirection connects various commands together, referred as pipelines. With this functioning, output for any specific command directly gets into the standard input of another one. Multiple commands are often used to increase efficiency. Some of the popular commands used in pipelines along with their functions are mentioned below.

Is-It | head- it can display a minimum 10 recent files in the current directory.

du | sort -nr - This command can display the entire list of the directory along with the space allocation to them. It helps sort files from smallest to largest.

find. -type f -print | wc -l - This command can display the number of files in the current directory along with the subdirectories in them

Filters- In pipelines, filters refer to a program that is mostly used in it, and they take standard input to operate based on five lines and then pass the standard output.

Performing task

- Linux provides a program known as LPR which can easily accept standard input and pass it to the printer. It is often used with the filters and pipes.